

MATERIAL SAFETY DATA SHEET

1.PRODUCT AND SAFETY INDETIFICATION

Product Name : PTS MB -109

Application

Water-extendable metalworking fluid/lubricant concentre which is normally to be diluted in water prior to use (typical dilution 3-10% in water). Refer to the supplier for further advice on suitability and dilution recommendations for specific applications.

2. COMAPOSITRION/INFORMATION ON INGREDIENTS

Contains highly refined mineral base oils, emulsifiers and corrosion inhibitors, with coupling agents and additional performance additives.

Note: the above components may not necessarily constitute the complete composition of the product.

3. HAZARDS IDENTIFICATION

Health and Safety

The undiluted product is strongly irritating in the eye with a potential to cause cornel injury if treatment is not prompt. In contact with the skin, the undiluted product may cause irritation which could become more intense if not promptly removed or if contact is frequent or prolonged. Prolonged or repeated contact with over strength emulsions may lead to defatting of the skin and/or slight irritation. For further information, refer to Section 11.

Environmental

The product contains mineral oil which will not readily biodegrade in anaerobic conditions and therefore can be environmentally persistent. For further information, refer to Section 12.

Special Hazards After Use

During use, metalworking emulsions may become contaminated, for example by metal particles and metal salts, other lubricants, and microbiological contaminants. These may increase the irritancy of the emulsions, and in some cases (e.g. contamination by chromium, cobalt and nickel) may be capable of inducing additional hazards. There is a possibility the small amounts of nitrosamines may be formed if sodium nitrite is added to the product.

4. FIRST-AID MEASURES

Eye Contact

Immediately wash eye thoroughly with plenty of clean water foe at least 15 minutes, ensuring eyelids are held open. For contact with undiluted fluid, obtain prompt medical attention. It is

advisable to obtain medical advice before removing contact lenses. For contact with diluted fluid, obtain medical attention if irritation or redness persists, or as an additional precaution.

Skin Contact

Undiluted product should be washed thoroughly from the skin with soap and water without delay. Remove contaminated clothing. Wash/laundry contaminated clothing before re-use. If irritation or other adverse skin reactions persist, obtain medical advice. If the use of metalworking emulsions gives rise to irritation or skin rashes, possible contamination and/or usage conditions may need to be investigated.

Inhalation

For effects produced by over-exposure, move to fresh air. If effects persist, obtain medical advice.
Ingestion

Do not induce vomiting. Wash out mouth with water and obtain medical attention. Milk or water drink may be beneficial (do not give anything to drink to an unconscious person). Treat symptomatically. If the product is aspirated into the lungs (e.g. during vomiting), send to hospital immediately. Show a copy of this data sheet to the doctor.

Notes for Doctors

Treat symptomatically – advisable not to induce vomiting due to the risk of aspiration and it is not usually necessary unless a large amount has been ingested or it has been contaminated with another product. Gastric lavage under supervised medical conditions can be carried out if necessary.

5. FIRE – FIGHTING MEASURES

Flammability

Not expected to be able to support combustion on account of its high water content. The residue left after water evaporation could be ignited by high energy sources. The diluted emulsions do not support combustion due to the high water content.

Flash Point:	>100c (not determinable)
Autoignition Temperature	> 150c (based on components)
Flammability Limits:	Not Established. Limits for mineral oil are in the range 1 -10%
Extinguishing Media:	None
Small Fires:	Foam, dry powder, carbon dioxide, sand or earth.
Large Fires:	Foam or water fog

Products of Combustion can produce a variety of compounds including

oxides of carbon, oxides of nitrogen; water vapour; unburnt hydrocarbons; partially oxidised organic compounds and other unidentified organic and inorganic compounds. Some of these compounds may be toxic.

Special Fire Hazards

Large surface areas exposed to air/oxygen (e.g. oil-soaked rags, paper or absorbed spillages) may be easily ignited and these should be cleared up at once.

Special Fire –Fighting Procedures

Firefighters should wear self-contained breathing apparatus. Do not spray water directly into storage containers due to boil over danger. Water may be used to cool nearby containers/surfaces. 6.

ACCIDENTAL RELEASE MEASURES

Contain spillage and prevent entry to drains or watercourse. Spillages can be slippery so affected areas should be thoroughly cleaned afterwards.

Safety Precautions

Wear suitable protective clothing, particularly eye protection. Refer to Section 8 for further details.

Small Spills

Soak in absorbent granules, sand or earth and collect solids into a suitable, marked container for proper disposal. Thoroughly clean spillage area as spillages can be slippery.

Large Spills

Bund the area using absorbent granules, booms, sand or earth. Temporarily seal exposed drainage outlets. Reclaim liquid directly or soak in an absorbent medium, and transfer to a suitable, marked container for proper disposal.

Disposal of Spillage

Disposal must be in accordance with local regulations and (in the UK) the Environmental Protection. Act 1990. Refer to section 13 further details.

7. HANDLING AND STORAGE

Handling

Avoid contact with eyes – wear suitable eye protection when handling the undiluted product. Avoid skin contact with the undiluted product. The use of appropriate barrier and after-work creams may be beneficial.

Storage

Store in dry conditions protected from frost and elevated temperature. Store in original containers or in other mild steel or high density polyethylene containers which are closable and clearly labelled. Certain requirements of the Control of pollution (Oil Storage) (England) Regulation 2001 may apply in England.

Additional Guidance

Metalworking fluids (MWFs) can create environmental, health and performance problems in use if not managed correctly – factors to be controlled include dilution, level of contamination, pH, fumes/misting, etc. The supplier can provide on dilution rates, and additional detailed advice on the control and maintenance aspects of MWFs.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure Limits

An occupational exposure limit for metalworking fluids (MWFs) has not been established. A control level of 1mg/m³ is suggested as being realistically achievable in most circumstances. The ACGIH exposure limit for mineral oil mist is 5mg/m³(8hr TWA)

Notes

Oil and MWF mist determination.

Primary Method: Gravimetric collection on a 5 u low ash filter. Fluorometric and IR techniques are also available for mineral oil mists.

Secondary Method: Detector tubes are available for mineral oil mist. The product contains significant proportions of the following components which have published.

Occupational Exposure Limits: 20-40%: Mineral oil (limits for mists or aerosols) General controls

General ventilation, safe working procedures and training should form the basis for exposure controls. Local forced extraction may be needed if mists, fumes or vapours are generated. Wash hands after use, before eating, drinking, or smoking, and before and after using the toilet. Contaminated clothing should be removed and laundered before re-use.

Personal Protective Equipment Eyes/Face

Eye protection is strongly advised when handling the undiluted product or if there is a risk of splashing with the diluted product. Chemical eye shield, spectacles or goggles can be used

Hands/Skin

Impervious gloves are recommended when handling the undiluted product. Prolonged or repeated contact with diluted metalworking fluid emulsions is often unavoidable the use of appropriate skin protective and reconditioning creams may be beneficial, and gloves should be considered whenever their use is practical and safe. Gloves should not have knitted wrists and/or open backs. PVC, nitrile or neoprene having a breakthrough time >360 minutes against alkanolamines and hydrocarbons, or which are suitable for use with synthetic metalworking fluids can be used. Latex and butyl rubber not recommended. Consider mechanical/ tear resistance if handling items which could damage the glove.

Respiratory Protection

Respiratory protection is not normally required. However, suitable respiratory equipment may need to be provided for those operations which generate vapour, mists or fumes and where exposure cannot be adequately controlled by local exhaust ventilation or other means. Respiratory half-masks (valved) giving protection against water based mists and particulates can be used Other EN345 safety boots (or En347 working shoes) resistant to oils and hydrocarbons. Work overalls to protect against skin contact

Environmental Controls

Suitable system design or appropriate controls should be in place to ensure that the product cannot discharge to drain, unless it is suitably treated to conform with local regulatory discharge standards.

NOTE : The above advice is based on and limited to our knowledge and experience of the product. It is the responsibility of the user to determine what particular controls and type of protective equipment are suitable and appropriate in relation to the specific conditions under which the product is used.

9. PHYSICAL AND CHEMICAL PROPERTIES

The following are indicative values only.

Appearance and State: Brown transparent Fluid

Odour: Mild characteristic

Flash Point: > 100c (Not determinable)

Autoignition Temperature: >250 c (based on components)

Flammability limits (% in air) : Not Established. Limits for mineral oil are in the range 1-10%.

Relative Density (@20 C): 0.89-0.91

Boiling Point/Range (C):>100(based on components, contains water)

Pour Point /Melting Point (C):<0

Vapour Pressure: Water is the most significant volatile component

Vapour Density (air=1): Not Applicable

Evaporation Rate (but. acetate =1) : <1

Kinematic Viscosity (@ 40o C) : No specific data. All hydrocarbon components are significantly > cSt.

Acidity/Alkalinity : Slightly alkaline

pH :8.9-9.1 @ 5% emulsion in water

Solubility In Water : Miscible to form translucent emulsion

Solubility in Solvents : Petroleum solvents

Water/ Oil partition Coefficient : No specific, test data. Coefficient for mineral oil is greater than 3.

Does not contain : Phosphorus, Chlorine, Phenol, Triazine Biocides ,Colouring Dyes

10. STABILITY AND REACTIVITY

Stability

This product is stable and unlikely to react in a hazardous manner under normal conditions of use.

Conditions to Avoid Extremes of temperature (preferably, store between 5 and 30 C). Protect from frost. Do not store above 60 C for prolonged periods in contact with aluminium-containing materials as there is a small possibility. In certain circumstances, that alkanolamines from compounds in the product could react with aluminium to release hydrogen gas.

Materials to Avoid

Strong oxidising agents (e.g. chlorates, peroxides) ; strong acids products containing sodium nitrite. The product may soften some rubbers and other incompatible elastomeric sealing materials. Do not store in containers made from copper, aluminium or zinc.

Decomposition Product

Thermal decomposition can produce a variety of compounds, the nature of which will largely depend on the conditions bringing about decomposition. Incomplete combustion or thermal

decomposition may be expected to generate such materials as particulate matter an unburnt hydrocarbons ; oxides of carbon; ammonia ; oxides of nitrogen ; oxides of boron; water vapour; partially oxidised organic compounds; and other unidentified organic and inorganic compounds.

Toxicological data is based on information on compounds and knowledge and experience of this and similar products.

11. TOXICOLOGICAL INFORMATION

Acute Toxicity

Ingestion Oral LD50: >2000 (MG/ Kg rats) The product is expected to have low order of acute oral toxicity – ingestion is not regarded as significant health hazard likely to arise in normal use.

Swallowing significant quantities may cause discomfort, nausea, irritation of digestive tract, and diarrhoea. aspiration into the lungs caused by vomiting or regurgitation following ingestion can be hazardous with possible resultant chemically induced pneumonia.

Dermal Dermal LD50:>2000 (mg/Kg rabbits) Dermal toxicity is not regarded as a health hazard likely to arise in normal use – prolonged skin contact is unlikely to result in the absorption of harmful amounts.

Inhalation Inhalation LC50: Not Established/No data Due to its low volatility , the product is unlikely to give rise to vapours which would present a significant inhalation hazard at ambient temperatures. High temperatures or atomising systems may lead to generation of vapours, mists or fumes which could cause irritation to eyes and respiratory tract, and pulmonary irritation.

Corrosivity /Irritation Eyes

Eye contact with the undiluted product may cause strong irritation and stinging .Then may be potential to cause corneal injury if treatment is not prompt. Dilute emulsions are expected to cause only slight transient irritation or redness.

Skin

The undiluted product in brief or occasional contact with intact skin can cause slight irritation which may become more intense if not promptly removed or if the skin is abraded or cut .Prepared emulsions are surface active and slightly alkaline, and prolonged or repeated contact with them, especially if the emulsions are over strength, may cause defecting of the skin, slight irritation and dermatitis.

Respiratory

High temperature or atomising systems may give rise to vapours, mists or fumes which could irritate eyes and respiratory tract.

Sensitisation

Contains biocide. This may be capable of causing allergic skin reactions in susceptible individuals,or it may cause individuals who are already sensitized to react to its presence in the product.

See also Section 16.

Chronic Toxicity

Repeated exposures to high concentrations of oil mists may cause a chronic inflammatory reaction of the lungs and give rise to a benign form of pulmonary fibrosis. This risk can be avoided by ensuring proper controls to minimise exposure to mists and fumes within the suggested controls limits (see Section 8).

Carcinogenicity

No carcinogenic effects are anticipated with this type of products during normal use. All mineral oils incorporated in the product have been highly refined. However, if sodium nitrite (or other nitrosatable material) is added to the product, there is a possibility that, over a period of time , this could react with diethanolamine or diethanolamine compounds in the product to form small amounts of potentially carcinogenic nitrosamines.

Mutagenicity

There are no reports of mutagenic effects attributable to the use of this type product.

Reproductive Toxicity

There are no reports of reproductive effects attributable to the use of this type of product.

Additional Notes Contamination and degradation of emulsions during use can affect the above properties, and in some cases may introduce additional hazards.

12. ECOLOGICAL INFORMATION

Ecological data is based on information on components and knowledge and experience of this or similar products

Mobility

The product will disperse as an emulsion in water. If released on land, small quantities will be absorbed in the upper soil layers where biodegradation may take place. Larger quantities may penetrate into anaerobic soil layers where mineral oil and some other organic compounds may persist. Many of the components have a high soil absorption coefficient which should help to prevent significant contamination of ground water. If it reaches the water table, the mineral oil could disperse as an emulsifiers in the products have also penetrated the soil layers.

Degradability and Persistence

The individual components range from readily to slowly biodegradable. The product contains mineral oil which has limited ready biodegradability when tested by methods CEC L-33-A-93 and OECD 301B. Mineral oil will biodegrade slowly in aerobic water and sediments and is considered to be ultimately biodegradable but it can be persistent in anaerobic conditions. Mineral oil loading can impair the functioning of sewage treatment plants.

Bioaccumulative Potential

The product will disperse as an emulsion in water, and some components will solubilise in water .Mineral oil has potential to bioaccumulate –its physical properties and slow rate of bio- degradation

suggest that mineral oil could interfere with the normal functioning of ecological cycles, and a contaminated area could be slow to recover.

Aquatic Toxicity

The product is not expected to be highly toxic to aquatic life. Mineral oil is not considered toxic to aquatic life (LC50>100MG/L), but has a potential to bioaccumulate. If released to water, the product will disperse as an emulsion and may deplete the oxygen supply to bottom dwelling organisms. There is a possibility that small amounts of nitrosamines may be formed with the nitrogen content in the water or in the presence of nitrites. The product contains a small amount of boron, water – soluble borates are widely distributed naturally in the soil and sea. Boron is an essential micronutrient for plants – but it is phytotoxic in higher concentrations.

Additional Notes

The product consists of components of low volatility which are not expected to be significant quantities **WGK Classification:** 2 (VwVwS Annex 4)

13. DISPOSAL CONSIDERATIONS

Do not contaminate ponds, water courses, soil or drains. Ensure all means of disposal comply with national and regional regulations. These include: in the EC:-the Waste Framework Directive (75/442/EEC and 87/101/EEC); and in the UK :- the Environmental Protection. Act 1990, Environmental Act 1990, Hazardous Waste Regulations 2005 and amendments.

Note: It is the end –user's responsibility to determine the regulatory status of waste

Undiluted Product

Do not dispose of untreated waste down the drains. The product may be incinerated in suitable equipment capable of handling water –containing mixtures, and under controlled conditions. Alternatively, the product can be disposed of via an authorised person/licensed waste disposal contractor. The product can also be diluted and treated as a diluted emulsion as below.

Diluted Product

Do not dispose of untreated emulsions down the drains. Dispose via an authorised person /licensed waste disposal contractor. Alternatively, emulsions can be treated in an appropriate effluent treatment facility (e.g. chemical splitting or ultrafiltration) to separate mineral oil and other components from the water phase. The clarified water phase may contain dissolved salts, surfactants, trace hydrocarbons, and other dissolved materials. It should not be discharged to drain without the approval of the appropriate local authority and without checking for compliance with issued consent conditions or other requirements. Further treatment may be required. The non-aqueous phase can be disposed of as for the undiluted product.

14. TRANSPORT INFORMATION

ADR	Not Regulated
Sea Transport	Not regulated
Marine Pollutant	No
Air transport	Not Regulated
UN classification	Not Regulated
UN number	Not Regulated
UN pack Group	Not Applicable

15. Administrative INFORMATION

Classification Irritant	Irritant
ANSI Signal Word	CAUTION
Label Information	Irritating to eyes and skin.

Avoid contact with skin and eyes. In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Wear suitable gloves. This material and its container must be disposed of as hazardous waste. Note: The above classification applies to the undiluted product as supplied. It may not apply if the product is diluted for use. If used in formulating other products, the formulations will need to be re-assesses in order to establish any applicable hazard labelling requirements.

16. OTHER INFORMATION

HMIS@ Rating

Health : 1
Flammability : 1
Physical Hazard : 0
Personal Protection : Assigned by the user

NFPA Rating

Health : 1
Flammability : 1
Reactivity : 0
Special Hazards : ALK

For extra data with respect to explicit applications, allude to the Technical Data Sheet or contact the provider.

The data on this Data Sheet relates just to the assigned item when utilized for the reasons showed.

PRODUCT DATA SHEET

PTS MB 109 (Hybrid Mineral Based Metal Working Fluid)

PTS MB 109 is petroleum oil base soluble cutting oil. The special petroleum oil neither polymerizes nor forms sticky mass so it lasts longer and the water dilution does not thicken the solution. IT is formulated with E.P. additives to give good lubricity. So you get better surface finish, enhanced tool life **PTS MB 109** contains biocides to enhance sump life (minimum sump life is 4 weeks or more).

Advantage

- (1) The water emulsion is stable. No separation in two layers or creaming on top.
- (2) Longer tool life due to high lubricity.
- (3) Good surface finish.
- (4) Longer sump life, biocide protected.
- (5) No rusting of machine or worked parts.
- (6) The base oil does not evaporate as it evaporates, fast in cheaper cutting oils, made from diesel cut, kerosene etc.
- (7) It is economical to use.

Specification:

Colour : Yellowish brown
Viscosity : 20+4 Second (ford cup -4)
Density : 0.90+0.05

Corrosion

Steel : Nil
Yellow metals : 0 to 1
Aluminium : Nil

Dilution

Cutting : 20 to 40
Grinding : 30 to 60

Why to use PTS MB 109 and not use cheaper cutting oils Some of our regular customers now, resisted to pay high price for **PTS MB 109** initially, while they were paying Rs.30/liter for cutting being used by them. We gave 20 liter of **PTS MB 109** . After a month we met, discussed and calculated liters of **PTS MB 109** used against liters of cheaper cutting oil used previously.

- Tool life
Surface finish
Sump life
- Rusting
Foul smell
Production loss

Over all working life oil . After collecting data from all users and analyzing, it was found that.

- 1)Cheaper cutting oils due to their low boiling points evaporate fast
- 2) lubrication was not good
- 3)Gave lower sump life
- 4) Required more quantity addition daily to keep sump level

Don not provide corrosion protection while PTS MB 109 having high boiling point and very good lubricity it provided

- 1) Enhanced tool life
- 2) Better surface finish
- 3) longer sump life
- 4) Required minimum oil to keep sump level
- 5) Provided an oil layer for effective corrosion protection

* Cheaper cutting oils do not form good emulsions, so full advantage of using cutting oil is not achieved. While PTS MB 109 forms stable micro emulsion, does not separate or cream out. This micro emulsion offer best lubrication and penetrates between tool and job, so you get better surface finish, enhanced tool life and rust protection.

*Biocide Protection:- The cheaper oil does not contain biocide or maximum, they contain phenol as biocide. So the bacteria and fungi spoil cutting oil, develop foul smell and breaks emulsion. The worker complains about bad smell and you have to discard whole sump. Clean it again and remake the dilution. You add cleaning and sump remaking costs and production loss

*When we calculated higher sump life, better finish, enhanced tool life, corrosion prevention and losses in production and labour charges for cleaning sump and recharging it, PTS MB 109 was found economical and cheaper oils were found costly. So it is clear that is all the way economical when you compare with cheaper oils.

Packing:- 20 Ltr & 210 Ltr.